

Nicolás Alonso Mella Ortiz

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Fifth-year Computer Science Engineering student with experience in software development, production deployment, and data analysis. Focused on the digitalization and automation of operational processes, as well as the integration of AI into real-world systems.

EXPERIENCE

Pinturas Panoramica

Dec. 2025 – Feb. 2026

Software Development Intern

Lautaro, Chile

- Developed a **full-stack web platform** using React, Tailwind CSS, Express.js, and PostgreSQL.
- Deployed **warehouse management** (WMS), transportation (TMS), and risk prevention form modules to production, **replacing untraceable paper and Excel workflows**.
- Built an **authentication and role-based access control** (RBAC) system by department using Passport.js, along with dashboards for **operational KPI tracking** using Shadcn and production management for an internal website.
- Designed a minimum viable product (MVP) to modernize legacy systems and establish the foundation for **plant floor control digitalization**.

PROJECTS

Academic Schedule Platform

- Replaced scattered **schedule lookups** across Excel files, PDFs, and slow systems with a centralized platform for the entire campus, featuring filters by course, classroom, and instructor.
- Used MongoDB to increase document/JSON read performance and Docker Compose to standardize and automate on-premise server deployment.
- Presented as a paper at **Congreso SOCHEDI 2025**.

QR Attendance System

- Designed an E2E platform to replace paper-based attendance tracking for tutoring sessions and teaching assistantships, featuring **real-time QR generation and validation**.
- Selected FastAPI to accelerate backend prototyping and ensure high performance during fast QR code validation.

Academic Mobility Dashboard

- Developed a functional demo of a student and faculty **mobility KPI dashboard** for the Center for Innovation and Teaching Development, complementing the design of research project indicators.
- Implemented with Django REST due to its robustness, built-in security, and ease of use.

EDUCATION

Autonomous University of Chile

Mar. 2022 – Present

B.S. in Computer Science Engineering

Temuco, Chile

SKILLS

Languages : Python, JavaScript, TypeScript, Java, Kotlin, SQL, HTML/CSS.

Frameworks & Web : React, Express.js, FastAPI, Django REST, Tailwind CSS.

Data & ML : Pandas, NumPy, Scikit-learn, Keras, Matplotlib.

Databases : PostgreSQL, MySQL, MongoDB.

Tools & DevOps : Git, GitHub, Docker, CI/CD, Linux, Claude Code, OpenCode.

Soft Skills : Teamwork, problem-solving, adaptability, effective communication.

Languages : Spanish (Native), English (Intermediate).